

MIT 18.01SC Final Exam Solutions

Complete Step-by-Step Guide

Problem 1: Derivatives

(a) Compute $f'(x)$ where $f(x) = x^3 e^x$.

Using the product rule, $\frac{d}{dx}[u \cdot v] = u'v + uv'$:

$$f'(x) = (3x^2)(e^x) + (x^3)(e^x) = x^2 e^x (3 + x)$$

(b) Compute $f^{(7)}(x)$ where $f(x) = \sin(2x)$.

Derivatives of sine repeat in cycles of 4 ($\sin \rightarrow \cos \rightarrow -\sin \rightarrow -\cos \rightarrow \sin$). Since $7 = 4 + 3$, the 7th derivative takes the form of the 3rd derivative ($-\cos$). Each derivative multiplies by the chain rule factor of 2.

$$f^{(7)}(x) = -2^7 \cos(2x) = -128 \cos(2x)$$

Problem 2: Tangents and Implicit Differentiation

(a) Find the tangent line to $y = 3x^2 - 5x + 2$ at $x = 2$.

First, find the y-coordinate: $y(2) = 3(2)^2 - 5(2) + 2 = 12 - 10 + 2 = 4$.

Find the slope: $y' = 6x - 5$. At $x = 2$, $m = y'(2) = 6(2) - 5 = 7$.

Using point-slope form: $y - 4 = 7(x - 2) \implies y = 7x - 10$.

(b) Show the curve $xy^3 + x^3y = 4$ has no horizontal tangent.

A horizontal tangent means $y' = 0$. Differentiating implicitly:

$$(y^3 + 3xy^2y') + (3x^2y + x^3y') = 0$$

Set $y' = 0 \implies y^3 + 3x^2y = 0 \implies y(y^2 + 3x^2) = 0$.

This implies either $y = 0$ or $y^2 + 3x^2 = 0$ (which only happens at $x = 0, y = 0$).

If $y = 0$, plugging into the original equation gives $0 = 4$, which is a contradiction. Thus, no horizontal tangent exists.

Problem 3: Limits and Derivative Definitions

(a) Compute $\frac{d}{dx} \left(\frac{x}{x+1} \right)$ using the definition of the derivative.

$$\begin{aligned} \lim_{h \rightarrow 0} \frac{\frac{x+h}{x+h+1} - \frac{x}{x+1}}{h} &= \lim_{h \rightarrow 0} \frac{(x+h)(x+1) - x(x+h+1)}{h(x+h+1)(x+1)} \\ &= \lim_{h \rightarrow 0} \frac{x^2 + x + hx + h - (x^2 + hx + x)}{h(x+h+1)(x+1)} = \lim_{h \rightarrow 0} \frac{h}{h(x+h+1)(x+1)} \\ &= \lim_{h \rightarrow 0} \frac{1}{(x+h+1)(x+1)} = \frac{1}{(x+1)^2} \end{aligned}$$

(b) Compute $\lim_{x \rightarrow \sqrt{3}} \frac{\tan^{-1}(x) - \pi/3}{x - \sqrt{3}}$.

This is the formal definition of the derivative of $f(x) = \tan^{-1}(x)$ evaluated at $x = \sqrt{3}$.

$$f'(x) = \frac{1}{1+x^2} \implies f'(\sqrt{3}) = \frac{1}{1+(\sqrt{3})^2} = \frac{1}{1+3} = \frac{1}{4}$$

Problem 4: Curve Sketching

Sketch $y = \frac{x}{x^2+1}$.

- **Symmetry:** $f(-x) = -f(x)$, so it is an odd function.
- **Asymptotes:** $\lim_{x \rightarrow \pm\infty} \frac{x}{x^2+1} = 0$, so horizontal asymptote at $y = 0$.
- **Extrema:** $y' = \frac{1(x^2+1) - x(2x)}{(x^2+1)^2} = \frac{1-x^2}{(x^2+1)^2}$. Set $y' = 0 \implies x = \pm 1$.
Local Min at $x = -1, y = -1/2$. Local Max at $x = 1, y = 1/2$.
- **Inflection Points:** $y'' = \frac{-2x(x^2+1)^2 - (1-x^2)(2)(x^2+1)(2x)}{(x^2+1)^4} = \frac{2x(x^2-3)}{(x^2+1)^3}$.
Set $y'' = 0 \implies x = 0, \pm\sqrt{3}$. Inflection points at $(0, 0)$, $(\sqrt{3}, \frac{\sqrt{3}}{4})$, and $(-\sqrt{3}, -\frac{\sqrt{3}}{4})$.

Problem 5: Optimization

Let the total width be x and height be y . The printed area is $(x-4)(y-8) = 50$.

Thus, $y-8 = \frac{50}{x-4} \implies y = \frac{50}{x-4} + 8$.

Minimize paper area $A(x) = x \cdot y = x \left(\frac{50}{x-4} + 8 \right)$.

$$A'(x) = \left(\frac{50}{x-4} + 8 \right) + x \left(\frac{-50}{(x-4)^2} \right) = \frac{-200}{(x-4)^2} + 8$$

Set $A'(x) = 0 \implies \frac{200}{(x-4)^2} = 8 \implies (x-4)^2 = 25$. Since $x > 4$, $x-4 = 5 \implies x = 9$.

Plug $x = 9$ into y : $y = \frac{50}{5} + 8 = 18$.

Dimensions: 9 inches wide by 18 inches tall.

Check for Minimum: $A''(x) = \frac{400}{(x-4)^3}$. For $x = 9$, $A''(9) > 0$, confirming it is a minimum.

Problem 6: Related Rates

Let x be the horizontal distance between the plane and the car. Total distance $D = \sqrt{x^2 + 1^2} \implies D^2 = x^2 + 1$.

Differentiating with respect to time t : $2D \frac{dD}{dt} = 2x \frac{dx}{dt} \implies D \frac{dD}{dt} = x \frac{dx}{dt}$.

At $D = 1.5$: $(1.5)^2 = x^2 + 1 \implies 2.25 = x^2 + 1 \implies x^2 = 1.25 = \frac{5}{4} \implies x = \frac{\sqrt{5}}{2} \approx 1.1$.

Given $\frac{dD}{dt} = -136$ mph:

$$(1.5)(-136) = \left(\frac{\sqrt{5}}{2} \right) \frac{dx}{dt} \implies -204 = (1.1) \frac{dx}{dt} \implies \frac{dx}{dt} \approx -185.45 \text{ mph}$$

This is the relative approach speed between plane and car. Let plane velocity $v_p = 120$. Since they are approaching each other along a straight line, $|v_c| + v_p = 185.45 \implies |v_c| = 185.45 - 120 = 65.45$ mph.

Problem 7: Limits with Integration

(a) $\lim_{n \rightarrow \infty} \sum_{i=1}^n \sqrt{1 + \frac{2i}{n} \frac{2}{n}}$.

This is a Riemann sum for $\int_0^2 \sqrt{1 + x} dx$.

$$\int_0^2 (1+x)^{1/2} dx = \left[\frac{2}{3} (1+x)^{3/2} \right]_0^2 = \frac{2}{3} (3^{3/2} - 1^{3/2}) = \frac{2}{3} (3\sqrt{3} - 1)$$

(b) $\lim_{h \rightarrow 0} \frac{1}{h} \int_2^{2+h} \sin(x^2) dx$.

By the Fundamental Theorem of Calculus, this is the derivative of $\int_2^x \sin(t^2) dt$ evaluated at $x = 2$, which is simply $\sin(2^2) = \sin(4)$.

Problem 8: Definite Integrals

(a) $\int_0^{\pi/4} \tan x \sec^2 x \, dx$. Let $u = \tan x \implies du = \sec^2 x \, dx$.

$$\int_0^1 u \, du = \left[\frac{1}{2} u^2 \right]_0^1 = \frac{1}{2}$$

(b) $\int_1^2 x \ln x \, dx$. Use integration by parts: $u = \ln x, dv = x \, dx \implies du = \frac{1}{x} dx, v = \frac{x^2}{2}$.

$$\left[\frac{x^2}{2} \ln x \right]_1^2 - \int_1^2 \frac{x^2}{2} \frac{1}{x} \, dx = (2 \ln 2 - 0) - \left[\frac{x^2}{4} \right]_1^2 = 2 \ln 2 - \left(1 - \frac{1}{4} \right) = 2 \ln 2 - \frac{3}{4}$$

Problem 9: Indefinite Integral

$\int \frac{x^2}{\sqrt{9-x^2}} \, dx$. Use trig substitution: $x = 3 \sin \theta \implies dx = 3 \cos \theta \, d\theta$. $\sqrt{9-x^2} = 3 \cos \theta$.

$$\int \frac{9 \sin^2 \theta}{3 \cos \theta} 3 \cos \theta \, d\theta = 9 \int \sin^2 \theta \, d\theta = \frac{9}{2} \int (1 - \cos 2\theta) \, d\theta = \frac{9}{2} \left(\theta - \frac{1}{2} \sin 2\theta \right) + C$$

Using $\theta = \sin^{-1}(x/3)$ and $\sin 2\theta = 2 \sin \theta \cos \theta = 2 \left(\frac{x}{3} \right) \left(\frac{\sqrt{9-x^2}}{3} \right)$:

$$= \frac{9}{2} \sin^{-1} \left(\frac{x}{3} \right) - \frac{x}{2} \sqrt{9-x^2} + C$$

Problem 10: Volume of Revolution

Use cylindrical shells. Radius a , hole diameter a (so bounds are $x = a/2$ to a).

$$V = \int_{a/2}^a 2\pi x \left(2\sqrt{a^2 - x^2} \right) dx = -2\pi \int_{a/2}^a -2x(a^2 - x^2)^{1/2} dx$$

Let $u = a^2 - x^2 \implies du = -2x \, dx$. Limits: $x = a/2 \rightarrow u = 3a^2/4$, $x = a \rightarrow u = 0$.

$$V = -2\pi \left[\frac{2}{3} u^{3/2} \right]_{3a^2/4}^0 = 2\pi \left(\frac{2}{3} \right) \left(\frac{3a^2}{4} \right)^{3/2} = \frac{4\pi}{3} \frac{3\sqrt{3}a^3}{8} = \frac{\pi\sqrt{3}}{2} a^3$$

Problem 11: Trapezoid Rule

$\int_1^5 \frac{e^x}{x} \, dx$ with 2 trapezoids. $\Delta x = \frac{5-1}{2} = 2$. Evaluate at $x_0 = 1, x_1 = 3, x_2 = 5$.

$$T_2 = \frac{\Delta x}{2} [f(x_0) + 2f(x_1) + f(x_2)] = \frac{2}{2} [2.7 + 2(6.7) + 29.7] = 2.7 + 13.4 + 29.7 = 45.8$$

Problem 12: Differential Equations

(a) $\frac{dm}{dt} = km \implies m(t) = m_0 e^{kt}$. Initial mass $m_0 = 100$.

Half-life is 1600: $50 = 100e^{1600k} \implies \ln(1/2) = 1600k \implies k = \frac{-\ln 2}{1600}$.

Formula: $m(t) = 100e^{-(\ln 2/1600)t} = 100(2)^{-t/1600}$.

(b) $m(1000) = 100(2)^{-1000/1600} = 100(2)^{-10/16} \approx 100(0.65) = 65$ mg.

Problem 13: Arc Length

$x = \int_0^t \cos(\pi u^2/2) du \implies \frac{dx}{dt} = \cos(\pi t^2/2)$.

$y = \int_0^t \sin(\pi u^2/2) du \implies \frac{dy}{dt} = \sin(\pi t^2/2)$.

Arc length $L = \int_0^{t_0} \sqrt{\left(\frac{dx}{dt} \right)^2 + \left(\frac{dy}{dt} \right)^2} dt = \int_0^{t_0} \sqrt{\cos^2(\dots) + \sin^2(\dots)} dt = \int_0^{t_0} 1 dt = t_0$.

Problem 14: Taylor Series

(a) $f(x) = \ln(1+x)$. $f(0) = 0, f'(0) = 1, f''(0) = -1, f'''(0) = 2$.

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots$$

(b) Ratio test: $\lim_{n \rightarrow \infty} \left| \frac{x^{n+1}}{n+1} \cdot \frac{n}{x^n} \right| = |x| < 1$. Radius of convergence is $R = 1$.

(c) $\ln(3/2) = \ln(1 + 1/2) \implies x = 1/2$. First two terms: $(1/2) - \frac{(1/2)^2}{2} = \frac{1}{2} - \frac{1}{8} = \frac{3}{8}$.

(d) Taylor's inequality: Error $|R_2(x)| \leq \frac{M}{3!}|x|^3$ where $M \geq |f'''(z)|$ on $[0, 1/2]$.
 $f'''(z) = \frac{2}{(1+z)^3}$. Max value on interval is at $z = 0$, so $M = 2$.

$$|R_2(1/2)| \leq \frac{2}{6} \left(\frac{1}{2} \right)^3 = \frac{1}{3} \cdot \frac{1}{8} = \frac{1}{24}$$

Problem 15: Bonus Proof

Prove $\frac{x}{1+x^2} < \tan^{-1}(x) < x$ for $x > 0$.

Part 1 ($\tan^{-1}(x) < x$): Let $f(x) = x - \tan^{-1}(x)$. $f(0) = 0$.

$f'(x) = 1 - \frac{1}{1+x^2} = \frac{x^2}{1+x^2} > 0$ for $x > 0$. Since $f'(x) > 0$, $f(x) > f(0) \implies x - \tan^{-1}(x) > 0$.

Part 2 ($\frac{x}{1+x^2} < \tan^{-1}(x)$): Let $g(x) = \tan^{-1}(x) - \frac{x}{1+x^2}$. $g(0) = 0$.

$$g'(x) = \frac{1}{1+x^2} - \frac{(1+x^2)(1) - x(2x)}{(1+x^2)^2} = \frac{1}{1+x^2} - \frac{1-x^2}{(1+x^2)^2} = \frac{(1+x^2) - (1-x^2)}{(1+x^2)^2} = \frac{2x^2}{(1+x^2)^2} > 0.$$

Since $g'(x) > 0$ for $x > 0$, $g(x) > g(0) \implies \tan^{-1}(x) - \frac{x}{1+x^2} > 0$. Proof complete.